GATE ME 90 Days Challenge

Day 52/90 · Practice Set with Detailed Solutions

SUBJECT	TODAY'S TOPIC	COUNTDOWN
Applied Thermodynamics	Turbomachinery	51 days to GATE

How to use this set: 15 questions (8 MCQ, 7 NAT) on impulse and reaction principles, velocity triangles, blade efficiency, compounding, degree of reaction and reheat factor in steam turbines; gas turbine expansion; the Pelton wheel, Francis and Kaplan turbines, specific speed, unit quantities, draft tubes and cavitation. Try them in **45 minutes** without looking at the solutions, then check the answer key and the step-by-step solutions. For NAT questions, enter the number; there is no negative marking for NAT in GATE. **Today's test:** open 7 to 9 PM in the **GATE ME 90 Days Challenge** course on our app and website (enrol once for ₹1: iurlk.courses.store/909124), 20 new questions on the same topic. Top scorers are announced in the next 8 AM challenge reel.

Questions

Q1 MCQ

A Pelton wheel is

(A) a tangential-flow impulse turbine

(B) an inward radial-flow reaction turbine

(C) an axial-flow reaction turbine

(D) a mixed-flow reaction turbine

Q2 NAT

The nozzle of a Pelton wheel works under a net head of 600 m with a coefficient of velocity of 0.97. Taking g = 9.81 m/s², the jet velocity (in m/s, round off to two decimal places) is _____

Q3 MCQ

The modern Francis turbine is best described as

(A) an axial-flow impulse turbine

(B) a mixed-flow (inward radial at entry, axial at exit) reaction turbine

(C) a tangential-flow impulse turbine

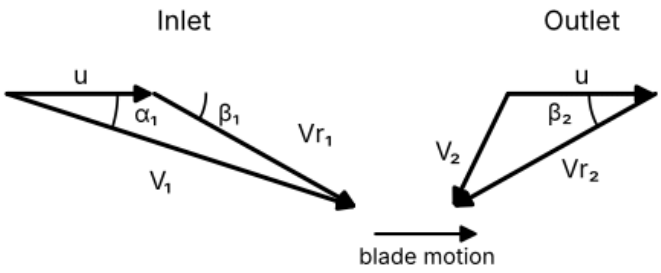
(D) an outward radial-flow impulse turbine

Q4 NAT

A hydraulic turbine develops 2000 kW under a net head of 40 m at 500 rpm. Its specific speed $N_s = N\sqrt{P}/H^{5/4}$ (N in rpm, P in kW, H in m) (round off to one decimal place) is _____

Q5 MCQ

The inlet and outlet velocity triangles of an impulse turbine blade are shown. If the blade is symmetrical ($\beta_1 = \beta_2$) and friction is neglected ($V_{r2} = V_{r1}$), then



(A) the blade efficiency is 100 %

(B) the whirl velocity at exit is always zero

(C) the axial thrust on the blades is zero, since $V_{f1} = V_{f2}$

(D) the relative velocity increases across the blade

Q6 NAT

Steam leaves the nozzle of a single-stage impulse turbine at 500 m/s at 18° to the plane of the wheel. The blade speed is 200 m/s, the blades are symmetrical and friction reduces the relative velocity at exit to 0.9 times that at inlet. The blade efficiency (in %, round off to two decimal places) is _____

Q7 MCQ

For a single-stage impulse turbine with nozzle angle α and symmetrical frictionless blades, the maximum blade efficiency and the optimum blade-speed ratio u/V_1 are

(A) cos α and cos α

(B) $\cos^2\alpha/2$ and cos α

(C) $2\cos^2\alpha/(1 + \cos^2\alpha)$ and cos α

(D) $\cos^2\alpha$ and (cos α)/2

Q8 NAT

A Kaplan turbine runner has a tip diameter of 4 m and a hub diameter of 1.6 m. If the velocity of flow through the runner is 7 m/s, the discharge (in m^3/s, round off to two decimal places) is _____

Q9 MCQ

The output of a Pelton wheel is governed (at constant head) mainly by

(A) a spear (needle) valve in the nozzle, with a jet deflector for sudden load rejection

(B) adjustable guide vanes (wicket gates)

(C) adjustable runner blades

(D) a draft-tube gate

Q10 NAT

A Pelton wheel delivers 2.5 MW of shaft power under a net head of 500 m with a discharge of 0.6 m^3/s. Taking g = 9.81 m/s^2, its overall efficiency (in %, round off to two decimal places) is _____

Q11 MCQ

In a multi-stage steam turbine the reheat factor is greater than one because

(A) the steam is reheated in the boiler between stages

(B) the friction losses of each stage reheat the steam, and part of this heat is recovered as work in later stages

(C) the blade speed increases from stage to stage

(D) the isentropic enthalpy drop is the same in all stages

Q12 NAT

In a 50 % reaction (Parsons) stage with symmetrical blading, the steam leaves the fixed blades at 240 m/s at 20° to the plane of the wheel, and the blade speed is 180 m/s. The work done per kg of steam (in kJ/kg, round off to two decimal places) is _____

Q13 MCQ

A reaction turbine works under a net head of 25 m. The atmospheric pressure head is 10.3 m of water and the vapour pressure head is 0.3 m. If the critical Thoma cavitation factor is 0.2, the maximum height of the runner exit above the tail-race is

(A) 10.0 m
(B) 15.0 m
(C) 5.0 m
(D) 2.0 m

Q14 NAT

Combustion gases (c_p = 1.15 kJ/kg·K, γ = 1.33) enter a gas turbine at 1100 K and expand through a pressure ratio of 6 with an isentropic efficiency of 85 %. The work output per kg of gas (in kJ/kg, round off to one decimal place) is _____

Q15 MCQ

For a hydraulic turbine working at different heads, the unit power P_u (power under unit head) is

(A) $P/\sqrt{H}$
(B) P/H
(C) $P/H^{5/4}$
(D) $P/H^{3/2}$

Answer Key

Q	Type	Answer
Q1	MCQ	(A) a tangential-flow impulse turbine
Q2	NAT	105.24 (accept 104.70 to 105.80)
Q3	MCQ	(B) a mixed-flow (inward radial at entry, axial at exit) reaction turbine
Q4	NAT	222.3 (accept 221.0 to 223.6)
Q5	MCQ	(C) the axial thrust on the blades is zero, since $V_{f1} = V_{f2}$
Q6	NAT	83.76 (accept 83.30 to 84.20)
Q7	MCQ	(D) $\cos^2\alpha$ and $(\cos\alpha)/2$
Q8	NAT	73.89 (accept 73.50 to 74.30)
Q9	MCQ	(A) a spear (needle) valve in the nozzle, with a jet deflector for sudden load rejection
Q10	NAT	84.95 (accept 84.50 to 85.40)
Q11	MCQ	(B) the friction losses of each stage reheat the steam, and part of this heat is recovered as work in later stages
Q12	NAT	48.79 (accept 48.54 to 49.04)
Q13	MCQ	(C) 5.0 m
Q14	NAT	385.9 (accept 384.0 to 387.8)
Q15	MCQ	(D) $P/H^{3/2}$

Detailed Solutions

Q1 **Answer: (A) a tangential-flow impulse turbine**

Concept: Classify turbines by the energy at inlet and the flow direction through the runner.

In a Pelton wheel all the head is converted to jet kinetic energy in the nozzle (impulse), and the jet strikes the buckets tangentially to the wheel: **(A)**.

Francis: (mixed/inward radial) reaction; Kaplan: axial-flow reaction.

Exam tip: Pelton wheels run in air at atmospheric pressure, so they need no draft tube and no casing pressure seal.

Q2 **Answer: 105.24 (accept 104.70 to 105.80)**

Concept: $V = C_v\sqrt{(2gH)}$.

$\sqrt{(2 \times 9.81 \times 600)} = \sqrt{11\,772} = 108.50$ m/s

$V = 0.97 \times 108.50 =$ **105.24 m/s**

Exam tip: For maximum efficiency the bucket speed is about $0.46V \approx 48$ m/s here.

Q3 Answer: (B) a mixed-flow (inward radial at entry, axial at exit) reaction turbine

Concept: Water enters the Francis runner radially inwards from the guide vanes and turns to leave axially into the draft tube.

Part of the head is still pressure head in the runner, so it is a reaction turbine: **(B)**.

It suits medium heads (about 40–300 m) and specific speeds of about 60–300.

Exam tip: Inward flow is self-regulating: if speed rises, the centrifugal head opposes the flow.

Q4 Answer: 222.3 (accept 221.0 to 223.6)

Concept: Specific speed is the speed of a geometrically similar turbine that develops 1 kW under 1 m head.

$\sqrt{2000} = 44.72$; $40^{1.25} = 100.6$

$N_s = 500 \times 44.72/100.6 =$ **222.3**

Exam tip: This lies in the Francis range; N_s below about 35 would point to a Pelton wheel.

Q5 Answer: (C) the axial thrust on the blades is zero, since $V_{f1} = V_{f2}$

Concept: Axial (flow) components: $V_{f1} = V_{r1} \sin \beta_1$, $V_{f2} = V_{r2} \sin \beta_2$.

With $V_{r2} = V_{r1}$ and $\beta_2 = \beta_1$, $V_{f2} = V_{f1}$.

Axial thrust $\propto \dot{m}(V_{f1} - V_{f2}) = 0$: **(C)**.

Exam tip: The exit whirl is zero only at one particular blade speed; the efficiency is limited to $\cos^2\alpha$ even then.

Q6 Answer: 83.76 (accept 83.30 to 84.20)

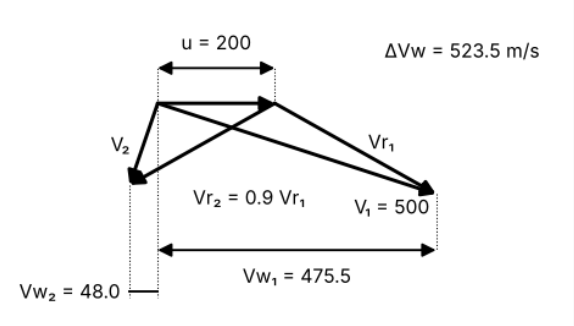
Concept: $\eta_b = 2u\Delta V_w/V_1^2$, with $\Delta V_w = V_{r1} \cos \beta_1 + V_{r2} \cos \beta_2$.

$V_{w1} = 500 \cos 18° = 475.53$ m/s; $V_{r1} \cos \beta_1 = 475.53 - 200 = 275.53$ m/s

$V_{r2} \cos \beta_2 = 0.9 \times 275.53 = 247.98$ m/s ($\beta_2 = \beta_1$)

$\Delta V_w = 523.51$ m/s; $\eta_b = 2 \times 200 \times 523.51/500^2 =$ **83.76 %**

Exam tip: Without friction the same stage gives 88.17 %: friction costs about 4.4 percentage points.



Q7 **Answer: (D) $\cos^2\alpha$ and $(\cos\alpha)/2$**

Concept: $\eta_b = 4\rho(\cos\alpha - \rho)$ with $\rho = u/V_1$ ($k = 1$, $\beta_1 = \beta_2$).

$d\eta/d\rho = 0 \rightarrow \rho_{opt} = \cos\alpha/2$

$\eta_{max} = 4\,(\cos\alpha/2)(\cos\alpha/2) =$ **$\cos^2\alpha$ (D)**

Exam tip: Option (C) is the 50 % reaction result, which occurs at $u/V_1 = \cos\alpha$.

Q8 **Answer: 73.89 (accept 73.50 to 74.30)**

Concept: The flow area of an axial runner is the annulus between hub and tip.

$A = (\pi/4)(D^2 - d^2) = (\pi/4)(16 - 2.56) = 10.556\ m^2$

$Q = A\,V_f = 10.556 \times 7 =$ **73.89 m^3/s**

Exam tip: Forgetting the hub gives 87.96 m^3/s; GATE often gives the hub as a fraction of the tip diameter.

Q9 **Answer: (A) a spear (needle) valve in the nozzle, with a jet deflector for sudden load rejection**

Concept: In an impulse turbine the flow is set by the jet area alone.

The spear moves in the nozzle to change the jet area and hence Q: **(A)**.

Closing the spear quickly would cause water hammer, so a deflector first turns the jet away from the buckets.

Exam tip: Francis: wicket gates; Kaplan: wicket gates plus runner-blade pitch (double regulation).

Q10 **Answer: 84.95 (accept 84.50 to 85.40)**

Concept: η_o = shaft power/water power, water power = ρgQH.

Water power = $1000 \times 9.81 \times 0.6 \times 500 = 2943$ kW

$\eta_o = 2500/2943 =$ **84.95 %**

Exam tip: $\eta_o = \eta_h \times \eta_m$ (× volumetric efficiency for reaction turbines).

Q11 **Answer: (B) the friction losses of each stage reheat the steam, and part of this heat is recovered as work in later stages**

Concept: Constant-pressure lines on the h–s chart diverge with increasing entropy.

Friction in a stage raises the entropy at its exit, so the next stage has a slightly larger isentropic drop: **(B)**.

Hence Σ(stage isentropic drops) > overall isentropic drop; RF = Σ/overall ≈ 1.03–1.05.

Exam tip: $\eta_{turbine} = RF \times \eta_{stage}$, so the turbine is a little more efficient than each stage.

Q12 **Answer: 48.79 (accept 48.54 to 49.04)**

Concept: Symmetric blading: $V_{r2} = V_1$ and $\beta_2 = \alpha_1$, so $w = u(2V_1\cos\alpha - u)$.

$V_1\cos\alpha = 240 \times 0.9397 = 225.53$ m/s

$\Delta V_w = 2 \times 225.53 - 180 = 271.05$ m/s

$w = 180 \times 271.05 = 48\,789$ J/kg = **48.79 kJ/kg**

Exam tip: At the optimum $u = V_1\cos\alpha$ the exit whirl is zero and $w = u^2$.

Q13 **Answer: (C) 5.0 m**

Concept: $\sigma = (H_{atm} - H_v - H_s)/H$; cavitation is avoided while $\sigma \geq \sigma_c$.

$H_{s,max} = H_{atm} - H_v - \sigma_c H = 10.3 - 0.3 - 0.2 \times 25$

$H_{s,max}$ = **5.0 m (C)**

Exam tip: High-specific-speed turbines (Kaplan) have large σ_c and often must be set below the tail-race.

Q14 **Answer: 385.9 (accept 384.0 to 387.8)**

Concept: $w = \eta_t c_p(T_1 - T_{2s})$, with $T_{2s} = T_1/r_p^{(\gamma-1)/\gamma}$.

$(\gamma - 1)/\gamma = 0.33/1.33 = 0.2481$; $6^{0.2481} = 1.5598$

$T_{2s} = 1100/1.5598 = 705.2$ K

$w = 0.85 \times 1.15 \times (1100 - 705.2)$ = **385.9 kJ/kg**

Exam tip: Use the γ and c_p given for the hot gas, not the air values (1.4, 1.005).

Q15 **Answer: (D) $P/H^{3/2}$**

Concept: For a given turbine, $V \propto \sqrt{H}$ and $Q \propto \sqrt{H}$, while $N \propto \sqrt{H}$.

$P \propto QH \propto H^{3/2}$, so $\mathbf{P_u = P/H^{3/2}}$ **(D)**.

Similarly $N_u = N/\sqrt{H}$ and $Q_u = Q/\sqrt{H}$.

Exam tip: Use unit quantities to predict the same turbine's speed, flow and power at a new head.

What next?

Day 52 test tonight, 7 to 9 PM

20 questions · 60 minutes · GATE marking (1/3 and 2/3 negative). Top 3 scorers are announced with name, marks and photo (if shared) in the next 8 AM challenge reel. Not enrolled yet? Join the ₹1 challenge course: iurlk.courses.store/909124

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